

Assignment 2: Experiment Comparison

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This assignment contained 4 experiments:

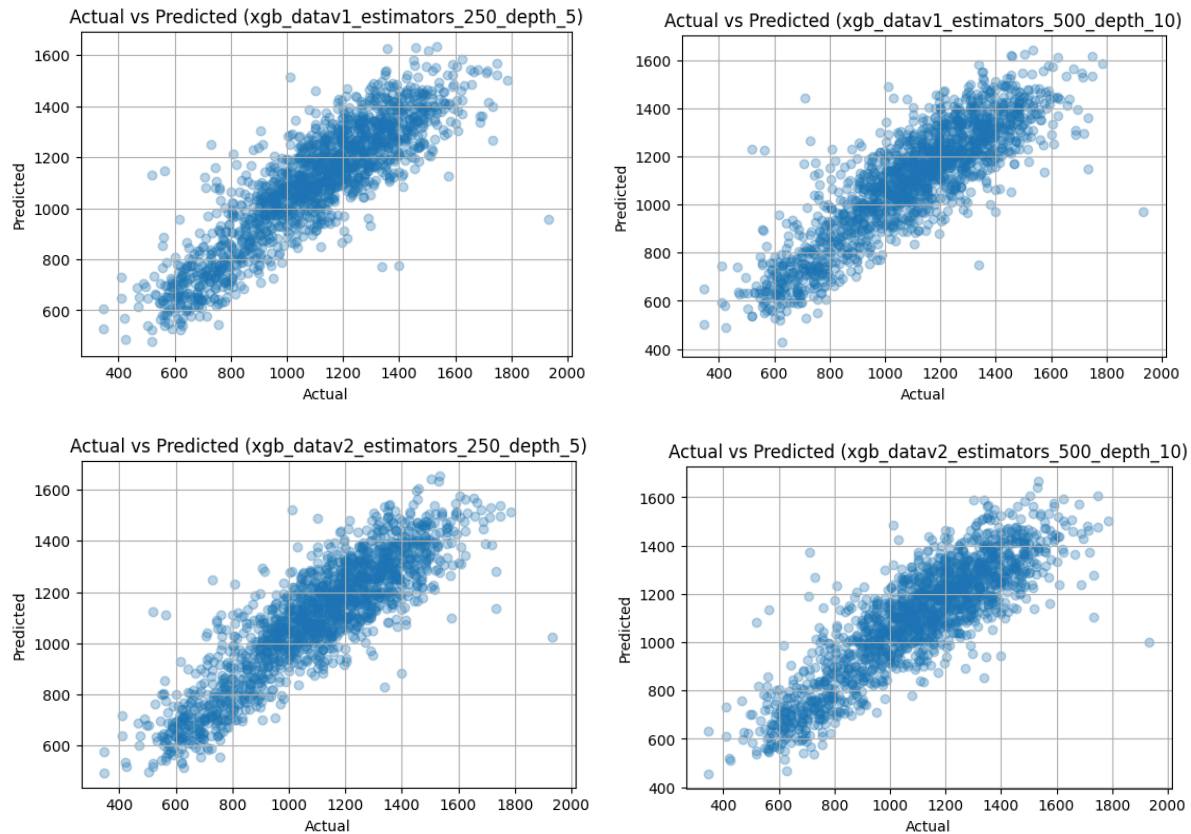
1. Data set version 1 (limited features) with hyperparameters $n_estimators = 250$ and $max_depth = 5$
2. Data set version 1 (limited features) with hyperparameters $n_estimators = 500$ and $max_depth = 10$
3. Data set version 2 (expanded features) with hyperparameters $n_estimators = 250$ and $max_depth = 5$
4. Data set version 2 (expanded features) with hyperparameters $n_estimators = 500$ and $max_depth = 10$

A priori, I expect experiment 3 to have the best evaluation metrics, because it has the most robust set of features and its hyperparameters allow for learning without encouraging overfitting (like the larger set of hyperparameters might do). However, it is possible that experiment 4 has more ideal hyperparameters and could outperform experiment number 3. Overall, I do not expect experiments 1 or 2 to perform the best, since they have fewer explanatory variables.

Now, let's evaluate my hypothesis and the results. All results are pulled from MLflow.

Experiment	R2	RMSE
1	.77	126.40
2	.74	134.24
3	.77	126.24
4	.74	133.60

We see that experiment 3 had the highest R2 and lowest RMSE, suggesting it was the best out of the four trials. Notably, though, it only barely outperformed experiment 1, which had the same hyperparameters but slightly fewer features. This suggests that the additional features created in data version 2 had a relatively limited benefit on the final model. Instead, the hyperparameter configuration appears to be more important to the results. See the four graphs below for a visual representation of model performance on actual vs predicted outcomes. These charts validate the numerical assessment that the models did not perform *too* differently from one another.



Screenshots from MLflow environment:

← healey_assignment_2_experiments Machine learning

MLflow 3 is available! Featuring unified ML and GenAI experiment tracking, improved model logging, prompt versioning, enhanced LLM judges, advanced tracing for end-to-end agent observability, and beyond. [Learn more about ML features](#) | [Learn more about GenAI features](#)

metrics.rmse < 1 and params.model = "tree" All time State: Active Datasets + New run

Sort: Created Columns Group by

	Run Name	Created	Dataset	Duration	Source
☐	xgb_datav2_estimators_500_depth_10	5 hours ago	-	0.9s	Assign
☐	xgb_datav2_estimators_250_depth_5	5 hours ago	-	2.0s	Assign
☐	xgb_datav1_estimators_500_depth_10	5 hours ago	-	0.8s	Assign
☐	xgb_datav1_estimators_250_depth_5	5 hours ago	-	1.1s	Assign

MLflow environment in DataBricks with model results

Run details

Description

[Add description](#)

Created at

Jul 18, 2026, 08:48 AM

Created by

healeyv@uchicago.edu

Status

Finished

Run ID

71a3f73c11e444e3a55c45bc2b78c6a1

Duration

0.9s

Dataset used

—

Type

Training

Tags

[experiment: databricks-colab-test](#)

Source

[Assignment 2 Healey](#)

Model registered

—

Artifacts

[View all](#)

prediction_plot_2_500_10.png

Metrics (2)

Metric	Latest	Min	Max
r2_score	0.74336686024916...	0.74336686024916...	0.74336686024916...
rmse	133.603026043621...	133.603026043621...	133.603026043621...

Parameters (4)

Parameter	Value
data feature version	2
max_depth	10
model	XGBoost
n_estimators	500

Specific model results (including artifacts, metrics, and parameters) in MLflow.